Uniklinikum Würzburg

The Hemibody Lab

Complementary antibody fragments (hemibodies) for effective targeting of breast cancer

Boris Nowotny, Mary Abena Attah, Laura Ceslik, Nina von Alten-Reuss, Vincent Borree, Gernot Stuhler

Department of Internal Medicine II, University Hospital Wuerzburg, Germany

BACKGROUND

Current T-cell re-directing strategies for cancer therapy are often associated with on-target/off-tumor effects, which can be life-threatening for the patient. The main reason for off-tumor reactivity is that most of the targeted receptors are only cancer associated and can also be found on healthy cells. To overcome this limitation, we have developed a new T cell-engaging antibody format (hemibodies) that comes in two complementary halves and addresses an antigen signature instead of a single antigen molecule. This empowers the hemibodies to target tumor cells more precisely, resulting in a reduction of off-tumor effects. For improving clinical usability, an Fc portion was added to the 1st generation hemibody format to increase serum half-life. For the treatment of breast cancer, we have generated a set of 2nd generation hemibodies targeting the receptors HER2, EpCAM, EGFR and ROR1.

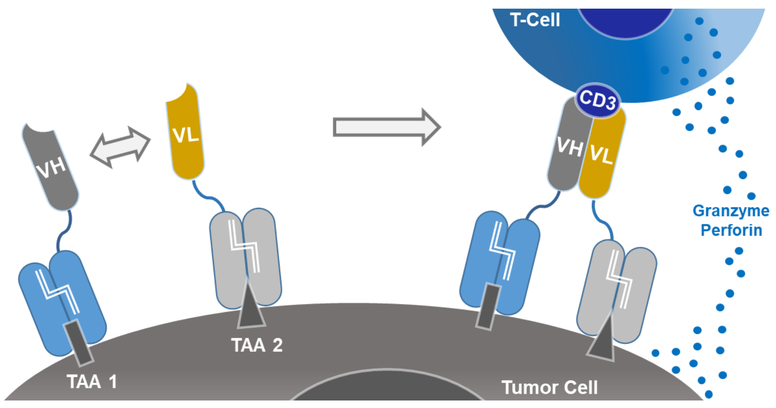

Figure 1: Hemibodies in action (1st generation format is shown). A hemibody consists of an antigen-binding domain (e.g. scFv, DARPin, VHH) fused to either the variable heavy (VH) or variable light (VL) chain domain of a T-cell activating anti-CD3 antibody. Upon binding to the respective target antigens, the complementary antibody halves come into close proximity and reconstitute a T-cell activating unit, followed by the formation of an immunological synapse and lysis of the target cell. TAA = tumor-associated antigen

HEMIBODY DESIGN AND MANUFACTURING

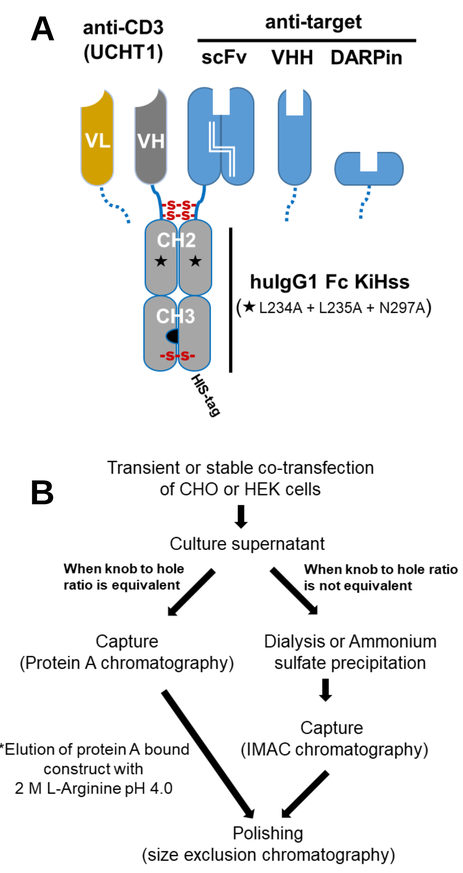

Figure 2: Architecture (**A**) and purification strategy (**B**) of KiH hemibody constructs (2nd generation format). ★ Point mutations that disrupt antibody effector function.

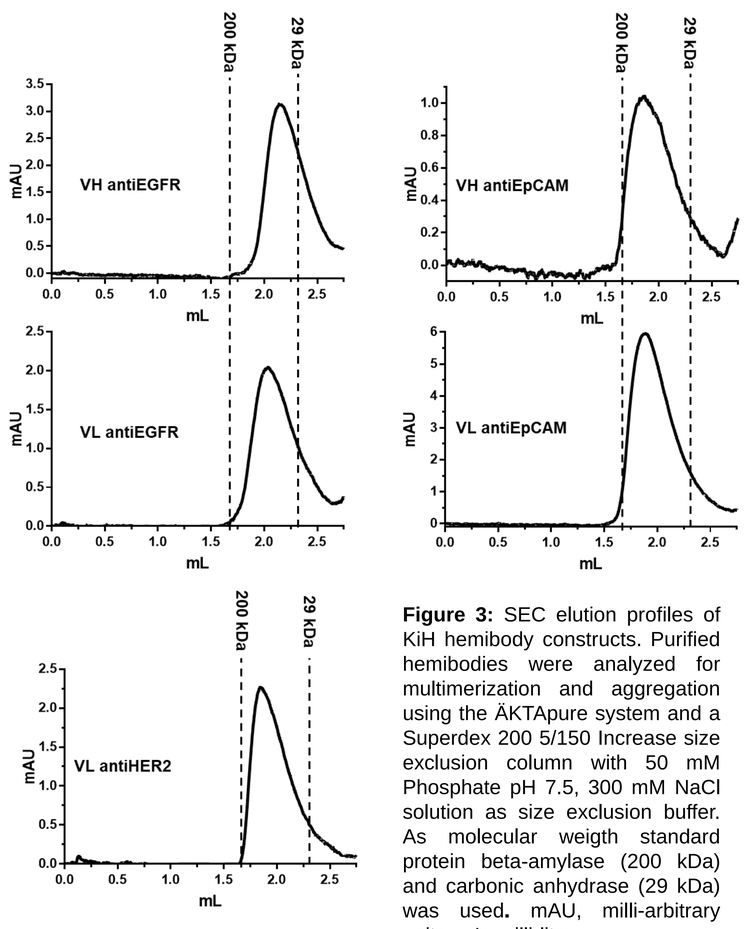

Figure 3: SEC elution profiles of KiH hemibody constructs. Purified hemibodies were analyzed for multimerization and aggregation using the ÄKTApure system and a Superdex 200 5/150 Increase size exclusion column with 50 mM Phosphate pH 7.5, 300 mM NaCl solution as size exclusion buffer. As molecular weigth standard protein beta-amylase (200 kDa) and carbonic anhydrase (29 kDa) was used. mAU, milli-arbitrary units; mL, milli-litre.

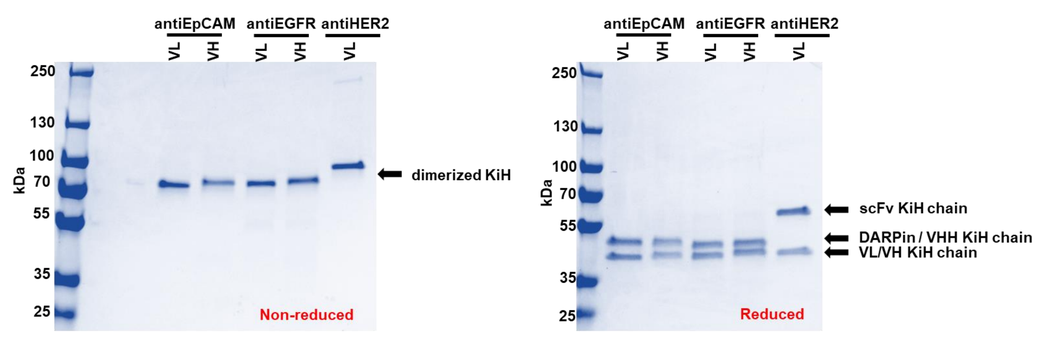

Figure 4: Purity of KiH hemibody constructs. Electrophoretic separation of SEC purified hemibody constructs. The separation was performed at non-reducing and reducing conditions using a 4-20% SDS-polyacrylamide gradient gel followed by Coomassie blue G250 staining. For analysis 0.5 µg hemibody construct was loaded per lane.

Table 1: KiH Hemibody construct properties and production yield.

Hemibody	Antigen binder format / clone	MW [kDa]	Yield [mg / L supernatant]	Purity [%]
VL antiEpCAM	DARPin / Ec4	82.4	1.29	95-100
VH antiEpCAM	DARPin / Ec4	83.9	0.57	95-100
VL antiEGFR	VHH / 9G8	79.1	0.32	95-100
VH antiEGFR	VHH / 9G8	80.6	0.14	95-100
VL antiHER2	scFv / Trastuzumab	90.9	0.16	95-100

HEMIBODY PERFORMANCE *IN VITRO*

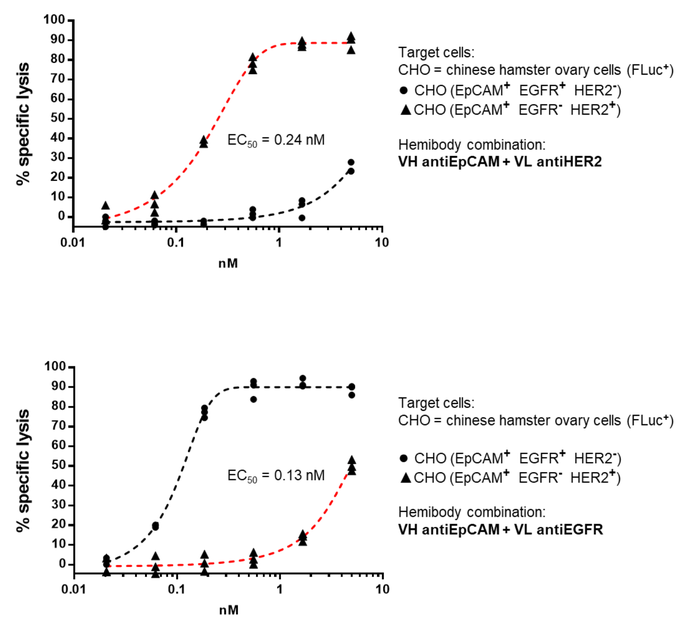

Figure 5: Hemibody mediated target-cell lysis. Dose dependency of target cell lysis was assayed with purified CD8 T-cells as effector cells and engineered CHO cells co-expressing the target antigens and firefly luciferase (FLuc) as target cells. nM, Nanomolar (hemibody concentration)

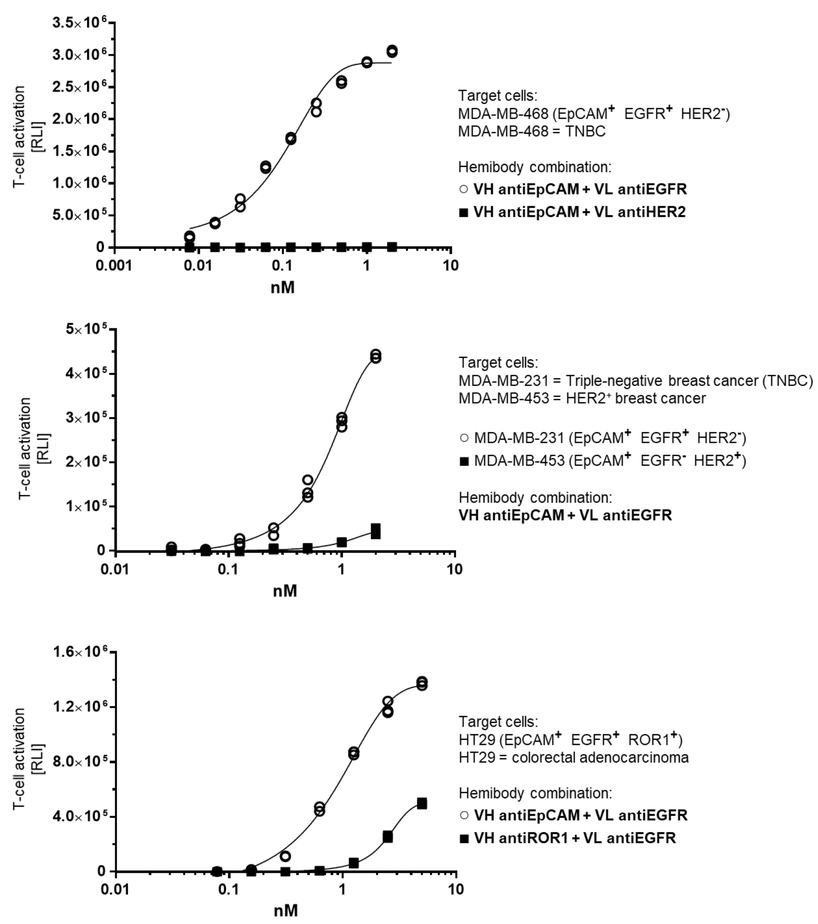

Figure 6: Hemibody mediated T-cell activation. Dose dependency of T-cell activation was assayed with Jurkat Lucia NFAT cells as effector cells and firefly luciferase expressing cancer cells as target cells. RLI, relative luminescence intensity; nM, Nanomolar (hemibody concentration).

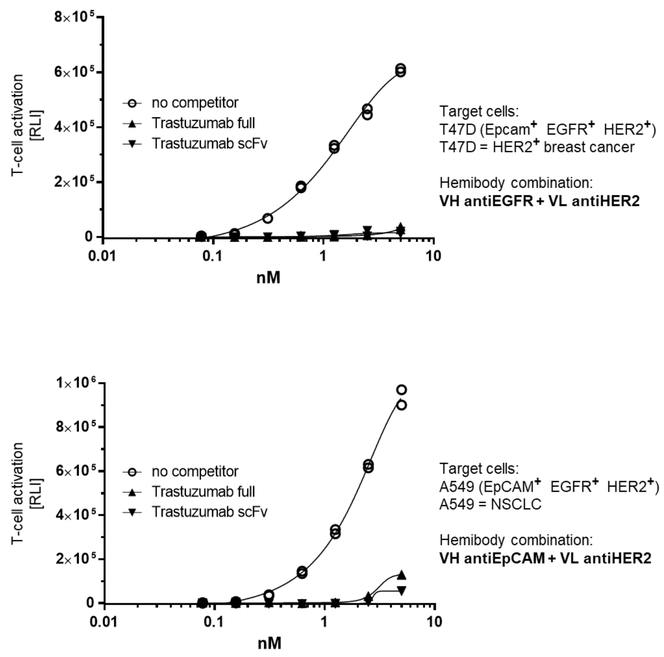

Figure 7: Hemibody mediated T-cell activation in the presence of a target-antigen competitor. Specificity of hemibody mediated T-cell activation was assayed with Jurkat Lucia NFAT cells as effector cells and firefly luciferase expressing cancer cells as target cells. For blocking the HER2 receptor we used Trastuzumab (antiHER2 antibody) in the full antibody and in the single chain variable fragment (scFv) format at a concentration of 100 nM. Two hours before addition of serial diluted hemibody constructs the effector and target cells were pre-incubated with Trastuzumab. RLI, relative luminescence intensity; nM, Nanomolar (hemibody concentration).

CONCLUSIONS AND OUTLOOK

We have succeeded in creating a new generation of hemibodies by fusing an Fc portion to improve serum half-life. These 2nd generation hemibodies precisely target breast cancer cells at picomolar dose. Next, we aim to develop a treatment strategy involving differential dosing combined with tumor-matching antigen combinatorics to address the inter- and intratumoral heterogeneity of breast cancer.